Strings

```
are any data under
'',  -> for a single character mostly
"", -> for a word(s) or text
""" """, -> for multi line strings
''' ''' -> same as above
```

Concatenate Strings

"" + "" + ''

Escape Sequences in Strings

- \n: new line
- \t: Tab means(8 spaces), move to the next tab stop
- \\: Back slash
- \': Single quote (')
- \": Double quote (")

String Formatting

| Old Style

```
formatted_string =  "%s %s" %(variable_name1, variable_name2)
```

%s -> string
%d -> integer
%f -> floating point number
%.numberofdigitsf -> float with fixed precision

| New Style

```
formatted_string = "{} {}".format(variable_name1, variable_name2)
```

add {.numberofdigitsf} for fixed precision

| String Interpolation / f - Strings

```
formatted_string = f"{variable_name1} {variable_name2}"
```

Python Strings as Sequences of Characters

a, b, c, d, e = "Jevon"
print(a) `#J`

| Access Characters by Index

```
name = "Jevon"
print(name[0]) #J
print(name[-5]) #J
```

| Slicing Python Strings

```
name[0:3]
# index 0 to 3 not include 3rd index

name[0:]
# start to the end

name[:3]
# from the first to 3rd
```

| Reverse Strings

```
[::-1]
```

| Skipping Characters

```
[0::2]
# Skip one character
```

String Methods

1. Capitalize

.capitalize() -> first character into capital

2. Count

.count(substring, start, end) -> return occurrences of substring in a string

3. Endswith

.endswith(substring) -> return True/Flase for a specified ending

4. Expandtabs

.expandtabs(number_of_spaces) -> replace tab or /t with spaces

5. Find

.find(substring) -> returns index of the first occurrence, else returns -1

6. Rfind

.rfind() -> same like find but the last occurenve

7. Index

.index(substring, start, end) -> similar to find but if not found returns valueError

8. Rindex

same like index but start from the last

9. Format

same like the new style formatting method

10. Isalnum

.isalnum() -> True/False to check alphanumeric

11. Isalpha

.isalpha() -> check alphabetic

12. Isdecimal

.isdecimal() -> check decimals (0 - 9)

13. Isdigit

.isdigit() -> check decimals and other unicode characters for numeric

14. Isnumeric

.isnumeric() -> check numbers or number related characters (more general than isdigit)

15. Isidentifier

.isidentifier() -> check if a string is a valid variable name

16. Islower

.islower() -> check all characters lowercase or not

17. Isupper

.isupper() -> check all characters uppercase or not

18. Join

" separator ".join() -> to concatenate strings

19. Strip

.strip(character) -> remove all given characters in a string

20. Replace

.replace(" ", " ") -> replace the first string with the second

21. Split

.split("separator") -> splits string by a separator

22. Title

.title() -> returns title cased string

23. Swapcase

.swapcase() -> converts uppercase to lowercase and the other way around

24. Startswith

.startswith(substring) -> check if a string starts with a specified substring